



Iheb Elleuch

B.Eng. Automation Systems and Production Informatics

Automation Engineering graduate specializing in industrial and hardware-near software development, visualization, and monitoring. Explored LLM-based solutions for writing process files in laser manufacturing with limited data.

Proven ability to bridge hardware and software domains to drive data-driven process optimization. Both self-driven and collaborative team player.

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EDUCATION

Bachelor of Engineering 09/2022 - 02/2026
Esslingen University of Applied Sciences (Göppingen, Germany)

Automation Systems and Production Informatics

- **Final Grade:** 1.9
- **Semester Abroad:** ESTACA, Paris, France (01/2025 - 05/2025)
- **Extracurriculars:** Co-founder of the underwater robotics initiative U.Stall

High School Diploma 09/2018 - 07/2022

Lycée Technique 9 Avril 1938 (Sfax, Tunisia)

- **Focus:** Mathematics | **Grade:** 2.0 (Equivalent)

PROFESSIONAL EXPERIENCE

Bachelor Thesis in Laser Technology 09/2025 - 02/2026

Bosch Manufacturing Solutions GmbH

- Demonstrated the feasibility of LLMs to assist in the creation of process files in industrial environments.
- Overcame limited data availability through synthetic data scaling techniques and the development of a benchmark method.
- Evaluated fine-tuning techniques and data curation methods to optimize performance on local hardware.

Research & Software Engineering Intern 09/2024 - 01/2025

Bosch Manufacturing Solutions GmbH

- Led the UI/UX development of a C#/WPF graphical interface for dynamic laser beam shaping.
- Designed a robust backend communication architecture using gRPC and Python.
- Expanded backend functionality to support new features and optimized laser beam shaping algorithms.
- Conducted routine validation of the software prototype.

Working Student - Real-Time Systems 12/2022 - 08/2024

Schuler Pressen GmbH (Andritz)

- Expanded the monitoring infrastructure and visualized critical data through Grafana dashboards.
- Collected and optimized data pipelines using Prometheus, PostgreSQL, and Node-RED routing.
- Developed and maintained time-saving automation scripts for faster data processing and easier upgrade iterations.

CORE COMPETENCIES

- **Programming:** C++, C#, Python, JavaScript, SQL, VBA
- **Technologies:** WPF, .NET, gRPC, Node-RED, Grafana, Prometheus, PostgreSQL, LLM Fine-tuning, Git (GitHub, GitLab, Azure)
- **Tools:** Git, Visual Studio, VS Code, Unity Engine, Axolotl, Llama-Factory, Office 365
- **Nationality:** German, Tunisian
- **Other:** Driver's License (Class B)

LANGUAGES

- **German:** Native
- **Arabic:** Native
- **English:** C1 Professional
- **French:** B2 Intermediate

KEY PROJECTS

- **U.Stall (MATE ROV):**
Lead of the software architecture. Development of system components for underwater robot control, communication, hardware integration, and competitive deployment. Participation: USA 2024 (Pioneer), USA 2025 (Explorer), Canada 2026 (Explorer).
- **Academic Robotics Projects:**
 - PinguBlocks:** Voice-controlled, penguin-shaped robot (Raspberry Pi, web interface).
 - KlemmbausteinIoT:** Modular robotics building platform with brick-like components and commercial hardware for academic teaching.